

COMMUNITY ENGAGEMENT PROJECT (CEP)

SKILL BRIDGE

Community Employment and Skill Matching Portal

Submitted in Partial Fulfilment

of the Requirements for the Award of the Degree

of

BACHELOR OF SCIENCE

[To be added: Information Technology / Data Science]

SEMESTER V

By

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Under the esteemed guidance of

[To be added: Guide name]

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Department of Computing

VSIT | Vidyalankar School of
Information Technology

RE-ACCREDITED : GRADE 'A' BY NAAC

DEPARTMENT OF COMPUTING

VIDYALANKAR SCHOOL OF INFORMATION TECHNOLOGY

(Autonomous College affiliated to University of Mumbai)

MUMBAI, 400037

MAHARASHTRA

2026 - 2027

VIDYALANKAR SCHOOL OF INFORMATION TECHNOLOGY

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CERTIFICATE

I hereby certify that Dhiraj Tendulkar (Roll Number: 24302A0004), student of Vidyalankar School of Information Technology, studying in T.Y. B.Sc. [To be added: Information Technology / Data Science], Semester V, has completed a C.E. project titled “Skill Bridge — Community Employment and Skill Matching Portal” at Khalsa College, Matunga, during the academic year 2026–27.

Internal Guide

Head of Department

Department of Computing

Internal Examiner

External Examiner

Date:



Principal

ORGANIZATION VISIT OBSERVATION SHEET

Attach the Organization Visit Observation Sheet duly signed by the reporting officer and mentor.

Organization: Khalsa College, Matunga

Student: Dhiraj Tendulkar | Roll Number: 24302A0004

[To be added: Signed observation sheet covering all three visits]

PROJECT GUIDE MEETING RECORD

Attach the Project Guide Meeting Record duly signed by the mentor.

Project: Skill Bridge — Community Employment and Skill Matching Portal

Student: Dhiraj Tendulkar | Roll Number: 24302A0004

[To be added: Signed guide meeting record]

DECLARATION

I, Dhiraj Tendulkar, a student of Vidyalankar School of Information Technology, T.Y. B.Sc. [To be added: Information Technology / Data Science], Semester V, declare that this report presents our Community Engagement Project, “Skill Bridge — Community Employment and Skill Matching Portal”, undertaken during the academic year 2026–27.

The fieldwork consisted of three visits to Khalsa College, Matunga. Our team collected real student responses by manually circulating a Google Form. The technical descriptions in this report concern the shared application developed for the project.

Sources used in preparing the report are identified in the references. Supporting forms and signatures are to be attached in their prescribed locations.

Signature of the Student with Date: _____

Dhiraj Tendulkar

Roll Number: 24302A0004

ACKNOWLEDGEMENT

I thank the University of Mumbai and Vidyalkar School of Information Technology for the opportunity to undertake this Community Engagement Project. It provided a setting in which classroom learning could be applied to a practical student-support problem.

I acknowledge our faculty mentor, [To be added: Guide name], and the support provided for the project. I also thank Khalsa College, Matunga, for the opportunity to carry out our three visits, and the students who took the time to respond to the Google Form.

I appreciate the shared effort of Vidhi Mahato, Gaurav Mhatre and Varun Patil throughout the team project. The report reflects a common body of fieldwork and software work, presented here in my own account.

ABSTRACT

Skill Bridge is a community employment and skill matching portal developed as part of our Community Engagement Project at Khalsa College, Matunga. Our team completed three visits to the college and gathered real student inputs by manually circulating a Google Form. The project addresses the need to connect opportunity information, student interests, skill preparation and placement follow-up in a clear workflow. Its objective is to provide useful student services while giving coordinators a separate workspace for operational decisions.

The application provides profiles, opportunity browsing, explained matching, saved opportunities, applications, workshop enrolment, feedback and guidance requests. Coordinator functions include managing listings, reviewing applications, maintaining training records and responding to requests. Administrative functions cover protected content, diagnostic queries and backup restoration. Public research views use aggregate information, while access to individual survey responses is restricted.

The implementation uses React, TypeScript, Vite, Tailwind CSS and Recharts for the frontend, with an Express server and SQLite persistence. Matching is a deterministic weighted process based on profile and opportunity fields. It returns reasons for its assessment and does not predict the probability of employment. Server-side authorization, input validation and database constraints support reliable operation.

The latest recorded verification passed 157 automated checks across acceptance, workflow, lifecycle and production-serving suites. The TypeScript check and production build also passed. These results establish the tested technical behaviour, including fresh production startup and administrator operations. They do not establish a measured improvement in placement outcomes. The portal provides a practical basis for organised placement support; verified survey totals, signed user acceptance and measured institutional impact remain to be added to the final evidence record.

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CHAPTER 1: INTRODUCTION

1.1 Project Background

Skill Bridge grew out of our Community Engagement Project at Khalsa College, Matunga. Our four-member team completed all three field visits at the same college. We gathered student inputs manually by circulating a Google Form. The project connects this requirement-gathering work with a web portal for placement opportunities, practical training, applications and guidance. In this report, I explain the shared project through its community purpose and the technical decisions used to implement it.

Table 1: Project team

Member	Roll number
Dhiraj Tendulkar	24302A0004
Vidhi Mahato	24302A0028
Gaurav Mhatre	24302A0067
Varun Patil	24302D0074

1.2 Problem Statement

The problem addressed is the difficulty of connecting a student's interests and skills with suitable placement opportunities while maintaining a clear record of applications and support needs. A listing alone does not explain whether a role fits a profile or what preparation is required. Coordinators also need a consistent way to review applications, manage workshops and respond to requests. Skill Bridge treats these activities as related parts of one process.

1.3 Need for the Project

The need for the project is a clearer connection between information and action. Students should be able to save an opening, submit an application and later see its status. Training and guidance should also have a recorded follow-up rather than ending at a request. Separate permissions make the portal easier to demonstrate and help protect student-specific records during routine use.

1.4 Objectives

The objectives link the community purpose to functions that can be checked in the application. I have separated the direct placement-support goals from the supporting requirements for usability and maintenance.

Primary Objectives

- Provide a distinct student workspace and coordinator workspace with server-enforced access.
- Support discovery, saving and application for opportunities with visible status updates.
- Connect workshop participation, feedback and guidance with coordinator follow-up.

Secondary Objectives

- Explain opportunity matching through understandable rules and reasons.
- Preserve related records and reject invalid or unauthorized changes.
- Provide aggregate research reporting while protecting individual responses.
- Maintain repeatable tests and practical operating instructions.

1.5 Scope

The implemented scope covers a local web application with student, placement coordinator and administrator access. It includes profiles, opportunity discovery, explained matching, applications, workshops, feedback, guidance, research summaries and administrative backup tools. The scope does not establish employer hiring decisions, actual placement gains or live institutional adoption. Hosting details and a signed handover record are [To be added].

1.6 SDG Alignment

The project aligns with SDG 4, Quality Education, through access to skill-development opportunities and clearer guidance. It also relates to SDG 8, Decent Work and Economic Growth, by helping students explore employment opportunities and prepare for relevant roles. These links describe the purpose of the intervention. They do not establish a measured change in educational or employment indicators.

CHAPTER 2: REQUIREMENT ANALYSIS AND LITERATURE REVIEW

2.1 Methodology Adopted

Our requirement-gathering method combined three visits to Khalsa College, Matunga, with manual distribution of a Google Form among students. The collected responses are real field inputs. The confirmed response total, collection dates and response summary are [To be added]. During development, requirements were translated into role-specific functions, implemented in stages and checked through builds, type checks and isolated tests. Software test counts are reported separately from field-survey evidence.

Table 2: Fieldwork record

Item	Confirmed information or record
Organization	Khalsa College, Matunga
Visits	Three visits, all at the same college
Collection method	Google Form manually circulated among students
Visit dates	[To be added]
Confirmed response count	[To be added]
Google Form link and response export	[To be added]
Verified findings and response summary	[To be added]

2.2 Stakeholder Interactions

Students were the respondents to the Google Form. The team members were Dhiraj Tendulkar, Vidhi Mahato, Gaurav Mhatre and Varun Patil. The software also models placement coordinators and administrators as operational stakeholders. Names, designations and signed records of any staff consultations are [To be added]. No individual staff statement is presented as an interview quotation without its supporting record.

2.3 Domain Overview

The domain combines campus placement support, student skill profiles and training coordination. An opportunity describes a role and its requirements; a student profile records interests and skills; an application connects the two. Workshops address preparation needs, while feedback and guidance provide routes for follow-up. Research reporting provides a separate view of student placement experiences.

2.4 Existing Process Analysis

The confirmed collection process used a Google Form circulated manually. That process gathers responses but does not itself provide student application histories, workshop capacity control or coordinator decision queues. Skill Bridge adds those operational functions. A detailed map of Khalsa College's existing placement procedures, including communication channels and responsible staff, is [To be added].

2.5 Existing Solutions and Related Work

The related-work discussion compares the collection method used in our project with the operational needs of the portal. Google Forms supported structured student input, while Skill Bridge provides persistent accounts and role-specific follow-up. No named commercial platform or research paper was supplied as an evaluated alternative. Verified literature citations and any formal comparative study are [To be added].

2.6 Technologies Studied

The selected stack is visible in the project implementation. React and TypeScript support the interface and shared data contracts; Express handles API requests; SQLite stores persistent records. Vite builds the frontend, Tailwind CSS supports styling and Recharts renders research charts. These are implementation choices, not evidence that other technologies were experimentally benchmarked.

Table 3: Technologies used

Technology	Role in Skill Bridge
React	Component-based frontend and role-specific pages
TypeScript	Typed models and compile-time checks for frontend and server
Vite	Frontend development server and production bundling

Technology	Role in Skill Bridge
Tailwind CSS	Interface layout and responsive styling
Recharts	Research charts and visual summaries
Node.js and Express	Runtime, API routes, authentication and authorization
SQLite through node:sqlite	Persistent relational records, foreign keys and transactions
tsx and oxlint	TypeScript execution and lint checks
Google Forms	Manual circulation and collection of real student field responses

2.7 Research Gap and Need for Proposed Solution

The practical gap is the connection between gathered information and a usable support workflow. Collecting student needs does not automatically provide relevant listings, application histories or staff follow-up. Skill Bridge addresses that gap through separate workspaces backed by a common database. A claim of academic novelty is outside the evidence available for this project.

2.8 Challenges Identified

The main technical challenges were preserving records during restoration, enforcing student–coordinator access boundaries and keeping validation consistent. Startup configuration and optional database values also required attention. The fieldwork challenges that can be quantified, such as incomplete submissions or participation levels, are [To be added]. These categories are kept separate so implementation defects are not misreported as community findings.

CHAPTER 3: PROPOSED SOLUTION

3.1 Solution Overview

Skill Bridge provides connected workspaces for students, placement coordinators and administrators. Students discover opportunities and request support; coordinators review and respond; administrators maintain protected configuration and recovery tools. This structure turns the project requirements into a sequence of recorded actions, with the server controlling permissions.

3.2 System Architecture

The browser runs the React interface and sends requests through the API service. Express receives these requests, resolves the session and checks the required role. Valid operations read or update SQLite, after which the result is returned to the interface. In production mode, Express also serves the built frontend assets. Environment configuration is loaded before database initialisation.

[To be added]

Figure 1: System Architecture of Skill Bridge (placeholder)

3.3 Process Flow Diagram

The principal flow begins with authentication and role identification. A student reviews relevant opportunities, checks the explanation and eligibility text, then saves or applies. The coordinator receives the application, records a decision and makes that status available to the student. Workshops and guidance follow similar request-and-response sequences.

[To be added]

Figure 2: Process Flow of Skill Bridge (placeholder)

3.4 Module Description

The modules share data where necessary but retain clear responsibilities. Authentication identifies the user, operational modules manage actions, and reporting summarises the relevant records. Administrative functions remain separate from ordinary student and coordinator work.

Table 4: Functional modules

Module	Responsibility
Authentication and access	Sessions, login/logout, role checks and account isolation
Student profile and matching	Profile preferences, skills and explained opportunity relevance
Opportunities and applications	Listing discovery, saved openings, submission and review status
Workshops	Session management, capacity, enrolment and attendance
Feedback and guidance	Student requests and coordinator responses
Research	Restricted response records and public aggregate reporting
Study and visit records	Three-visit documentation, action plans and evidence fields
Administration	Protected content, approved diagnostics and backup/restore

3.5 Advantages of Proposed Solution

The main advantage is continuity between student actions and coordinator responses. Explained matching adds context to opportunity discovery, and persistent records allow users to return to their work. Role-specific access reduces ambiguity during use. These are supported capabilities of the design; measured improvements in placement success or staff workload have not been established.

CHAPTER 4: SYSTEM DESIGN

4.1 System Design Overview

The design uses typed records for profiles, opportunities, applications, workshops and research responses. Relationships are stored in SQLite rather than reconstructed only in the interface. Validation checks input before persistence, while foreign keys and uniqueness constraints protect important relationships. The browser state presents these records but does not replace the database as the authoritative store.

4.2 Use Case Diagram

The use cases are organised around four actors: public visitor, student, placement coordinator and administrator. Public access is limited to appropriate information and aggregate research. A student works with personal records; staff operate management functions. The server checks these boundaries even when an endpoint is called directly.

Table 5: Actors and access

Actor	Permitted scope
Public visitor	Permitted public pages, opportunity information and aggregate research
Student	Own profile, saved listings, applications, enrolments, feedback and guidance
Placement Coordinator	Operational management, decisions, rosters and restricted research
Administrator	Protected configuration, diagnostics, backups and authorized management

[To be added]

Figure 3: Use Case Diagram of Skill Bridge (placeholder)

4.3 Workflow Diagram

The application workflow requires a valid student session, a selected opportunity and an accepted request. Duplicate applications are rejected. Coordinator review changes the stored application status, which is later displayed to the student. For workshop enrolment, the server also checks capacity and existing participation before committing the record.

[To be added]

Figure 4: Application and Workshop Workflow of Skill Bridge (placeholder)

4.4 Module Design

The module design connects frontend pages to shared context and API methods. Server endpoints operate on the corresponding database entities. The matching function is a deterministic weighted process: it compares profile information with listing fields and returns a score together with reasons. It is advisory and does not predict hiring probability.

Database records and relationships

Table 6: Database entities

Entity or group	Purpose
users; sessions	Account identities, roles and active authentication sessions
student_profiles	Student-specific skills, interests and academic information
opportunities; applications	Listings linked to student submissions and review decisions
saved_opportunities	Student-specific saved listings
training_sessions; workshop_enrollments	Workshop information and participation relationships
feedback_submissions; guidance_requests	Requests and staff follow-up
survey_responses	Structured responses with provenance and controlled access

Entity or group	Purpose
employer_outreach	Staff-managed employer contact records
system_documents; app_metadata	Content documents and database initialisation state

[To be added]

Figure 5: Key Components of the Skill Bridge Database (placeholder)

Opportunity matching process

Table 7: Matching score components

Component	Points	Implemented rule
Department	40	Awarded when the listing includes the profile department
Preferred role	25	Case-insensitive containment between a preferred role and listing title
Skill overlap	Up to 20	Seven points per overlapping listed skill, capped at 20
Location	10	Preferred location matches, or the listing uses remote work
Academic cutoff	5	Entered percentage meets a detected two-digit percentage cutoff

The score is the sum of the awarded components, bounded between 0 and 100. Skill points are calculated as $\min(20, 7 \times \text{number of matching listing skills})$. A score of 75 or more is labelled High Fit; 50–74 is Moderate Fit; 30–49 is Partial Fit; lower scores are General Opportunity.

Text comparisons use case-insensitive containment, so the result depends on the terms entered in the profile and listing. The eligibility check detects a two-digit percentage cutoff and returns an eligibility caution where appropriate. It is not a complete eligibility parser. Students must review the original criteria even when the fit score is high.

CHAPTER 5: PROJECT IMPLEMENTATION

5.1 Development Methodology

Development followed an iterative cycle of implementation, review and correction. The team maintained the existing application while addressing verified issues in permissions, restoration, validation and startup. Type checks, production builds and isolated tests were used after changes. This process allowed defects to be examined without resetting the real project database.

5.2 Module-wise Implementation

The source is organised into React pages, reusable components, shared context, API services and server modules. AppContext coordinates session and data refreshes. The API service sends authenticated requests, and Express performs role checks and validation. SQLite retains the resulting records. This structure implements the proposed separation between student actions, coordinator decisions and administrative maintenance.

Table 8: Implementation mapping

Source area	Implementation role
src/pages	User-facing pages
src/components	Reusable interface components
src/context/AppContext.tsx	Shared session and application state
src/services/apiService.ts	Client API communication
src/utils/matching.ts	Rule-based matching function
server/index.ts	API and frontend-serving entry point
server/db.ts	Schema and database initialisation
server/validation.ts	Input and restoration validators
server/config.ts	Environment and runtime configuration
scripts	Tests, provisioning and backup utilities
README.md; .env.example	Setup instructions and configuration examples

Authentication and data integrity

Passwords are hashed with scrypt and a salt. Session tokens are carried in an HttpOnly cookie, and protected routes verify the session and role. Public registration is restricted to the student role. Database uniqueness constraints help prevent duplicate applications and enrolments. Restore operations validate supported content and relationships before transactional updates.

Research summaries

The reporting process groups responses by the requested filters and computes counts, distributions and rating summaries. Non-rated responses such as “Have not attended” are not treated as numeric satisfaction scores. Public reporting uses aggregate endpoints; individual responses are limited to authorized staff. The actual field-response totals and numerical findings are [To be added].

5.3 User Interface Solution

The interface gives students and coordinators different navigation and actions. Students see personal application and support tools, while coordinators receive management views. Forms and status messages help users understand what happened after an action. Responsive styling is implemented, but a complete interactive mobile and accessibility walkthrough remains a separate verification item.

5.4 Screenshots of Developed Solution

The following locations are reserved for screenshots of the developed application. Each image should show the actual current interface and use test accounts with no private student information visible.

(add picture)

Figure 6: Student Workspace of Skill Bridge (placeholder)

(add picture)

Figure 7: Placement Coordinator Workspace of Skill Bridge (placeholder)

(add picture)

Figure 8: Research Summary View of Skill Bridge (placeholder)

CHAPTER 6: TESTING AND VALIDATION

6.1 Test Cases

Validation covered normal actions, invalid input and unauthorized requests. The workflow suite used a disposable SQLite database, allowing records to be created and removed without changing the real database. The representative cases below show how the tests relate to the project requirements.

Table 9: Representative test cases

Area	Scenario	Expected result
Authentication	Invalid credentials and unauthenticated requests	Rejected
Role authorization	Student requests staff-only resources	Rejected
Account isolation	One student requests another student's records	Private records not exposed
Application lifecycle	Submit, reject duplicate, record coordinator decision	Expected records and status retained
Workshop lifecycle	Enrol, reject duplicate, cancel, enforce capacity	Participation rules enforced
Research privacy	Public/student access to raw responses	Blocked; aggregate reporting available
Content and restoration	Malformed documents and invalid relationships	Rejected; valid restoration supported
Diagnostics	Client-supplied SQL and approved query identifiers	Raw SQL blocked; approved queries work
Production lifecycle	Fresh database, restart and admin provisioning	Startup and provisioning checks passed

6.2 Test Results

The latest run passed all four automated suites, the TypeScript check and the production build. Lint completed with warnings. These results apply to the tested behaviour; browser interaction and live-host verification remain separate.

Table 10: Recorded automated results

Suite	Checks	Result
Acceptance	37	Passed
Workflow and security	97	Passed
Production lifecycle	14	Passed
Production serving	9	Passed
Total	157	Passed

6.3 User Acceptance Testing

Signed UAT records, participant details and observed outcomes are [To be added]. Actual users should verify the intended tasks in the browser. Automated acceptance checks do not establish formal acceptance by the college.

6.4 Feedback from Stakeholders

The real Google Form response summary and approved participant quotations are [To be added]. Annexure D contains a blank reporting-officer form. No unsupported statement is attributed to a student or staff member.

6.5 Performance Evaluation

The tests establish functional behaviour, not measured speed or scale. Response times, throughput and concurrent-user capacity are [To be added]. A documented workload and target-host measurements are required before making performance claims.

CHAPTER 7: RESULTS AND IMPACT ASSESSMENT

7.1 Deliverables Submitted

The developed outputs include the portal source, SQLite-backed workflows, automated tests, configuration examples and project documentation. These are available project artefacts. The date and acknowledgement of formal submission or handover to Khalsa College are [To be added]. Photographs and signed records remain separate supporting attachments.

7.2 Results Achieved

The achieved technical result is a connected portal with protected student and staff workflows. The previous startup and schedule-validation failures were corrected, and the latest four-suite run passed. Production-mode checks verified serving, administrator authentication and an authorized update. These results describe software operation rather than a change in student placement outcomes.

7.3 Impact on Organization

The portal offers the college a possible way to organise placement-support activities and preserve their history. Students can obtain clearer next steps, while coordinators can work from structured queues. Actual adoption, workload reduction and student benefit must be confirmed through use and feedback. A measured organisational impact summary is [To be added].

7.4 SDG Contribution

The delivered features support the educational and employment-preparation aims of SDG 4 and SDG 8. Workshops and guidance address preparation, while opportunity discovery and application records support access to career information. The contribution is a practical digital support mechanism; verified long-term SDG outcomes are beyond the current evidence.

CHAPTER 8: CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

Skill Bridge connects the community project with a practical placement-support application. The three visits to Khalsa College, Matunga, and manual Google Form collection provide the fieldwork setting. The technical work establishes differentiated access, persistent workflows and a passing automated test baseline. Final photographs, signed records and user-acceptance evidence will complete the submission record.

8.2 Key Findings

The implementation shows that permission checks, validation and database relationships are central to a reliable portal. A successful build alone is insufficient; startup and workflow tests exposed problems that required correction. Matching explanations are also important because a numerical score needs context. Field-response findings must be drawn from the original response export.

8.3 Recommendations

Before institutional use, assign an operational owner, verify the deployment configuration and complete a student–coordinator browser walkthrough. Keep backups and test recovery periodically. Add the original Google Form summary, visit dates and signed attachments to the report. Review privacy requirements before publishing any research output.

8.4 Future Enhancements

Possible future work includes configurable matching rules, opt-in reminders and richer follow-up reporting. These would require additional implementation and evaluation. Accessibility testing and measured performance on the deployment host would also improve confidence. None of these proposed extensions is presented as an existing feature.

REFERENCES

Research Papers

[To be added: Verified research papers actually consulted, formatted in MLA style.]

Books Referred

[To be added: Books actually consulted, with verified MLA bibliographic details.]

Website

[To be added: Websites actually consulted and the topics referred to, with verified URLs and access details.]

Project source material:

Skill Bridge Project Team. Skill Bridge: Community Employment and Skill Matching Portal. Project source code, 2026.

Skill Bridge Project Team. "Project Setup and Verification." README.md, 2026.

Skill Bridge Project Team. "Opportunity Matching." src/utis/matching.ts, 2026.

Skill Bridge Project Team. "Survey Questionnaire." src/data/defaultSurveySeed.ts, 2026.

Skill Bridge Project Team. "Verification Suites." scripts/verify-acceptance.ts, scripts/test-workflows.ts, scripts/test-production-startup.ts, and scripts/verify-production-serving.ts, 2026.

ANNEXURES

The annexures retain the labels prescribed in the supplied format: A, B, C, D, G and H.

Annexure A: Field Visit Photographs

Visit 1 — Khalsa College, Matunga

Date of visit: [To be added]

(add picture)

Figure 9: Visit 1 at Khalsa College, Matunga

Visit 2 — Khalsa College, Matunga

Date of visit: [To be added]

(add picture)

Figure 10: Visit 2 at Khalsa College, Matunga

Visit 3 — Khalsa College, Matunga

Date of visit: [To be added]

(add picture)

Figure 11: Visit 3 at Khalsa College, Matunga

ANNEXURE B: SURVEY QUESTIONNAIRE

We collected real student inputs by manually circulating a Google Form. The questions below reproduce the questionnaire present in the project source. The original circulated form link and any differences from this version are [To be added].

1. Which department or course are you studying in?

Response options: Computer Science (CS); Information Technology (IT); Data Science (DS).

2. What is your current year of study?

Response options: First; Second; Third; Fourth; Other.

3. What job role or career domain are you mainly interested in?

Response options: Open text response.

4. How many campus recruitment drives offered roles relevant to your preferred domain during the current academic year?

Response options: 0; 1–2; 3 or more; Not sure.

5. Is the placement cell currently meeting your placement-related needs?

Response options: Yes; Partly; No; Not enough experience to judge.

6. How often do you receive placement information early enough to prepare and apply?

Response options: Always; Sometimes; Rarely; Never; Not applicable.

7. How useful has the placement training you attended been for your preferred role?

Response options: 1–5, where 1 is Not useful and 5 is Very useful; Have not attended.

8. What is the biggest barrier you face in getting a suitable placement?

Response options: Limited relevant roles; Eligibility restrictions; Skill gaps; Late information; Interview preparation; Location or pay mismatch; Other; No major barrier.

9. Which placement support do you need most urgently?

Response options: Relevant employer connections; Technical training; Aptitude preparation; Resume and interview help; Career guidance; Timely alerts; Internship support; Other.

10. How satisfied are you with overall placement support?

Response options: 1–5, where 1 is Very dissatisfied and 5 is Very satisfied; Not enough experience to judge.

11. Briefly describe your experience with the placement process.

Response options: Open text response.

12. What one change would most improve placement support for you?

Response options: Open text response.

ANNEXURE C: INTERVIEW QUESTIONS

The project includes an interview guide for placement-coordination discussions. The questions below reproduce that guide. Interview dates, respondent identity, consent and recorded answers are [To be added]. Illustrative answers in the software are not reproduced as actual staff statements.

1. How are student preferences and skills collected?
2. How many relevant employers and roles were available by department?
3. What explains differences in opportunities?
4. How are new employers approached?
5. What makes specialist employers difficult to attract?
6. How are notices received and published?
7. How are training needs identified?
8. How is training effectiveness measured?
9. What alternatives exist for students excluded by eligibility criteria?
10. Are duties, location, and pay details checked before publication?
11. What staffing, time, or budget constraints exist?
12. Which concerns are already being addressed?
13. What additional actions are feasible, with owners and timelines?
14. What can Skill Bridge support?
15. How is student feedback reviewed and answered?

ANNEXURE D: REPORTING OFFICER FEEDBACK FORM

Blank form for assessment of the prepared project

Project: Skill Bridge — Community Employment and Skill Matching Portal

Organization: Khalsa College, Matunga

Student: Dhiraj Tendulkar | Roll Number: 24302A0004

Reporting officer name: _____

Designation: _____

Date of review: _____

Relevance of the project to student/community needs

Usefulness and clarity of the developed solution

Observations on the demonstration

Suggested improvements

Overall assessment and acceptance status

Reporting officer signature: _____

Organization stamp, if required: _____

ANNEXURE G: SOURCE CODE REPOSITORY

Repository link: [To be added]

Submitted version or commit identifier: [To be added]

Contents uploaded and verified by: [To be added]

The following inventory identifies the project contents to include in the submission. It does not confirm that a repository has already been published.

Table 11: Repository contents

Source area	Contents
src/pages	User-facing pages
src/components	Reusable interface components
src/context/AppContext.tsx	Shared session and application state
src/services/apiService.ts	Client API communication
src/utils/matching.ts	Rule-based matching function
server/index.ts	API and frontend-serving entry point
server/db.ts	Schema and database initialisation
server/validation.ts	Input and restoration validators
server/config.ts	Environment and runtime configuration
scripts	Tests, provisioning and backup utilities
README.md; .env.example	Setup instructions and configuration examples

Exclude passwords, private environment files, real student records and local database backups from any public repository. Provide authorised access to sensitive evidence separately.

ANNEXURE H: USER MANUAL

This manual follows the principal tasks of each role. Use the account provided for the intended role and keep credentials private. The actual hosting URL is [To be added].

Student workflow

1. Sign in and review your profile, including department, skills and preferred roles.
2. Open opportunities, apply relevant filters and read the listing details.
3. Review matching reasons and the original eligibility requirements before applying.
4. Save an opening for later or submit one application; check its status in your workspace.
5. Use the workshop section to enrol, review participation or cancel where permitted.
6. Submit feedback or a guidance request and return to view the response. Sign out when finished.

Coordinator workflow

1. Sign in with a placement coordinator account.
2. Maintain opportunities and workshop records through the management views.
3. Review student applications and record decisions or feedback.
4. Use workshop rosters to maintain the available participation and attendance information.
5. Respond to feedback and guidance requests; review research records only for authorised work.
6. Confirm save messages and revisit the record if a request fails.

Administrator workflow

1. Use an explicitly provisioned administrator account.
2. Maintain protected content and run only the approved diagnostic queries.
3. Export a backup before a sensitive maintenance operation.
4. Restore only a validated backup and check the resulting records.
5. Keep database backups and credentials in protected storage.

Local setup and production preparation

Install a Node.js version compatible with the project dependencies and native SQLite support, then run npm install in the project folder.

Use `npm run dev` for local development. The configured API and frontend addresses should be read from the terminal output.

Run `npm run typecheck`, `npm run build` and `npm test` before preparing a release.

For production, set `NODE_ENV=production`, a persistent `DB_PATH` and `CLIENT_URL` matching the actual website origin. Configure HTTPS and the intended reverse proxy.

Provision an administrator with explicit credentials using `npm run admin:provision -- <username> <password>`. Use protected credential handling in the deployment environment.

Run `npm start` after the build. Verify login, authorised changes and direct route refreshes on the actual host.

Use `npm run backup:db` for the database backup utility. Test recovery with a disposable copy before relying on a backup. Do not run demo reset commands against the real database.

If an operation fails

Read the displayed error and check the current record before repeating a submission. Confirm that the server is running and the account has the required role. For repeated failures, record the action and error for the administrator without sharing private student data.